

HW1: Constrained control allocation for dynamic ship positioning using deep neural network

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Methodology of the code :

The neural network has an encoder-decoder architecture. The Layers of the encoder and decoder are LSTM layers primarily. We also have Dropout Layers between the LSTM layers. A tanh layer is specifically added to the encoder-decoder architecture in order to normalize the values received by the previous layer. This ensures that the values remain in the range specified

Data :

The data is user generated. We make a dataset of 1000000 entries for each F1, F2 , F3, alpha2, alpha3 for our particular use case. Using that we calculate the tau matrix which consists of thruster values i.e tau(sway) , tau(heave), tau(yaw).

The values for each of the 6 values are generated with the random walk function and the shape of the input to the encoder is 5 x 1000000. The dataset is further split into train- test of 80% and 20% respectively.

Loss Functions:

The model contains 5 loss functions L0, L1, L2, L3, L4, L5 which are combined to form the net loss function.

As mentioned in the documentation the weights for each loss are user generated and arbitrary. However we have manually assigned more weight to the MSE i.e L0 and MSE for allocated Force

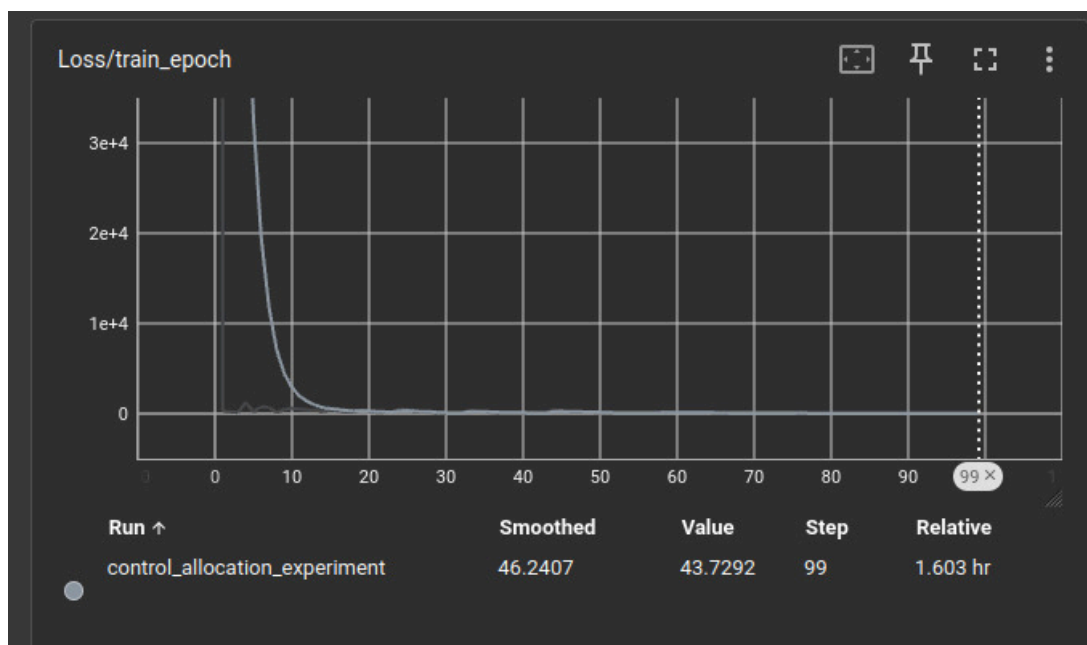
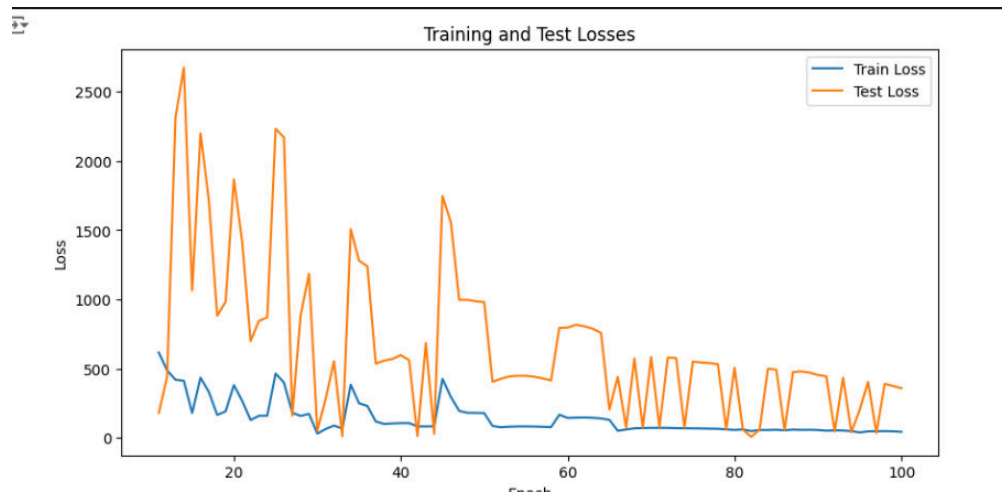
```
k = torch.tensor([1e0, 1e0, 1e-1, 1e-7, 1e-5, 1e-1]).to(device)
```

The average individual loss values we received for the model were

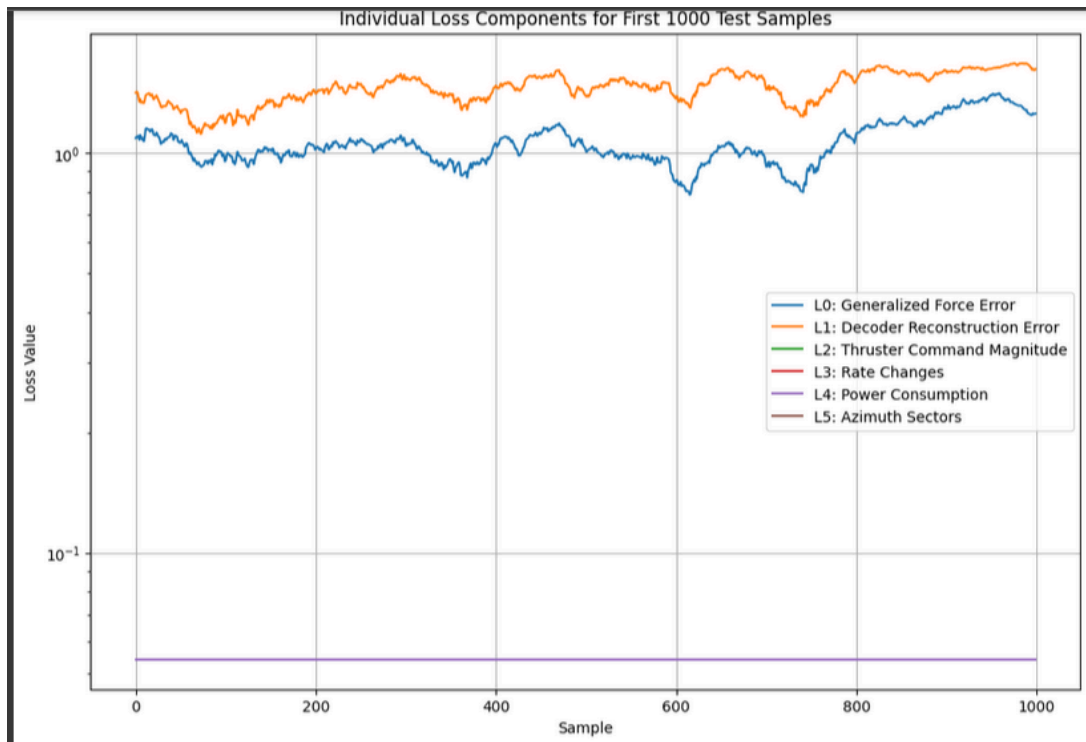
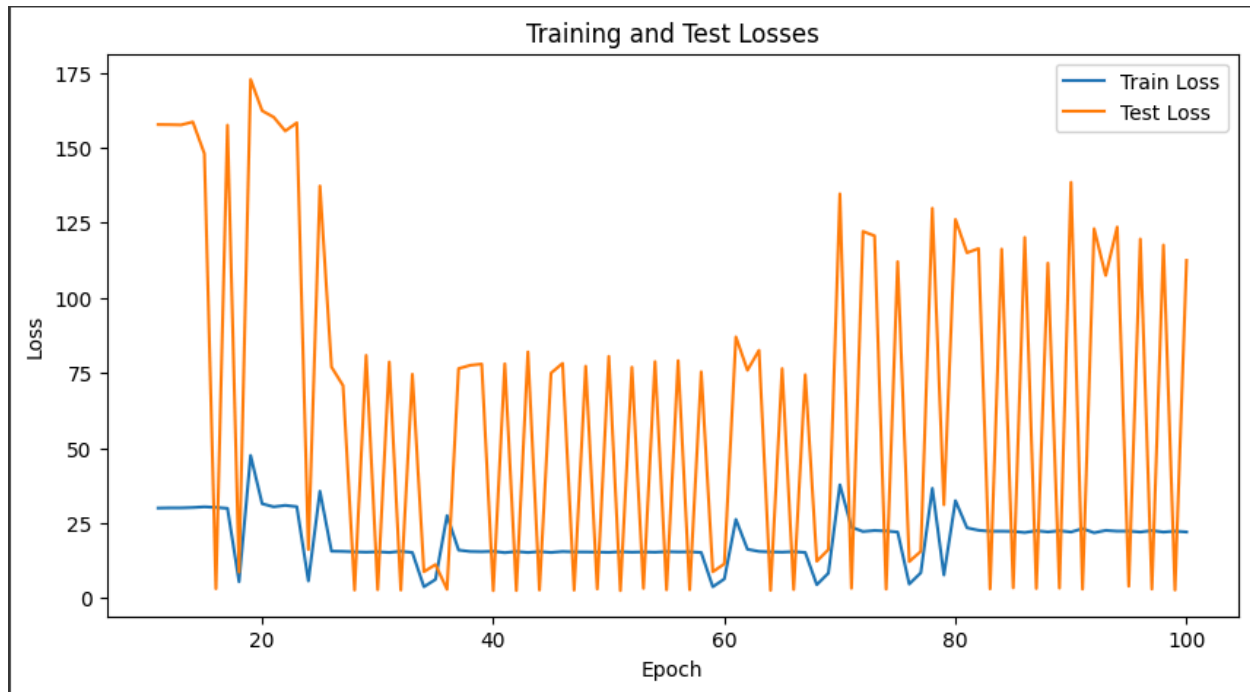
```
Average losses:
L0: 1.057355
L1: 1.459089
L2: 0.000000
L3: 0.000000
L4: 0.054132
L5: 0.000000
```

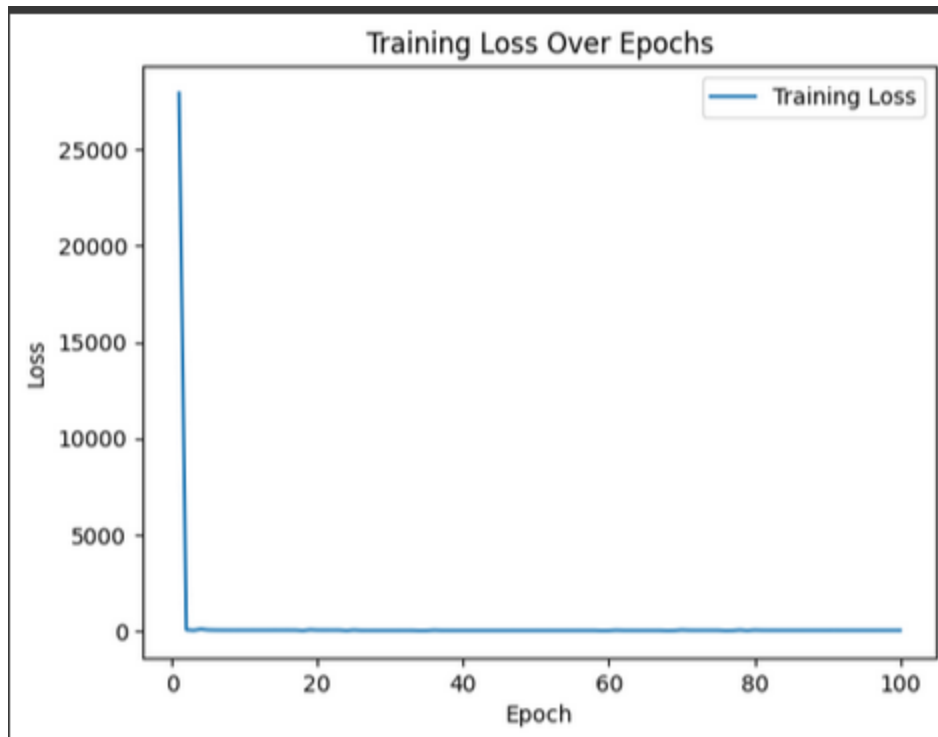
Where L3 = rate change penalty L4 = Power consumption minimization L5 = Sector Constraint Violation

Without Regularization :



With Regularisation and Dropout Layer





Due to regularization it is observed that the model starts oscillating and fails to capture the relation.

Hyperparameters:

- > The Hyperparameters here are the K values i.e the weights for the Loss function;
- > The Lambda for L2 regularization.
- > The learning rate for training
- > The dropout rate for LSTM layers